

Total No. of Questions : 8]

SEAT No. :

PE-2556

[Total No. of Pages : 2

[6583]- 85

T.E. (Electronics/E & TC)

DATABASE MANAGEMENT

(2019 Pattern) (Semester - V) (304183)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8. from following questions.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What are some common clauses used with SELECT query in SQL? Explain any three with example. [8]

b) What are different types of joins in SQL? Explain any one with suitable example. [6]

c) Define the use of TRUNCATE, DELETE and DROP statements in SQL? Explain any one with example. [4]

OR

Q2) a) Explain the use of constraints in SQL. List and explain any three constraints example. [8]

b) What is Cursor? Explain Implicit and Explicit cursor in PL/SQL with suitable example. [6]

c) Explain with example any two STRING functions used in SQL. [4]

Q3) a) When deadlock occurs? How to prevent them? How to recover if dead lock occurs. [6]

b) Define transaction. State and explain in brief the ACID properties of a transaction? [5]

c) Explain briefly different states of transaction with a neat state transition diagram. [6]

OR

P.T.O.

- Q4)** a) Explain Commit and Rollback operations of Transactions. [6]
b) What is Concurrent execution of transaction? Explain any two concurrency problems occurs during Concurrent execution of transaction with examples? [5]
c) Explain the term [6]
i) Conflict Serializability
ii) View Serializability

- Q5)** a) What are performance parameters for parallel database? [6]
b) Explain virtualization on Multicore processors with neat schematic diagram. [6]
c) What is Parallelism? Explain Inter query parallelism. [5]

OR

- Q6)** a) List the parallel database architectures and explain any one architecture. [6]
b) Explain Oracle Database Architecture. [6]
c) Explain client server architecture with suitable example. [5]

- Q7)** a) Explain Homogeneous and Heterogeneous Distributed Databases. [6]
b) Explain Fragmentation and Replication in Distributed data storage. [6]
c) What is distributed transaction? Explain failure modes of distributed transaction. [6]

OR

- Q8)** a) What is necessity of distributed database management system? What are the Factors Encouraging DDBMS. [6]
b) Explain two phase commit protocol in distributed database. [6]
c) Draw and explain client-server architecture for DDBMS. [6]

